claim	our assessment (an argument)	formal results it rests on
D answering	true in comparative form; not a new inference principle; survives the objection from statistical practice	**????????** and Props. 1.3, 1.4, A.1 and A.3
S logical form	descriptive part true of declarative textbook presentations (an observation); explanatory part open, and relative to a reasoner	§ 1.4; **????????**
E explanation	true in the unification sense, with at least four gaps	Props. 1.5 and A.5 and **????**
L the ladder	SAI is on step 2, with partial results on step 3; step 4 has no formal handle	**??????**
P progress	to be replaced by a list of formal differences; none concerns the shape of hypotheses in the idealised setting	Tab. 2

Table 1: The argument map: our assessments, which are arguments, and the results they rest on, which are theorems in their stated settings.

1 What a theory does and what it is

What does a scientific theory do? It may predict observations, the view behind Solomonoff induction (Solomonoff, 1964); answer questions posed in some language, why-questions included (van Fraassen, 1980, ch. 5); explain (Hempel and Oppenheim, 1948); or solve problems (Laudan, 1977). And what is it? A program that predicts observations; a set of sentences closed under consequence (Suppe, 1977); a class of models (Suppes, 1960; van Fraassen, 1980); a core of models with intended applications (Sneed, 1971; Balzer et al., 1987); or a practice of exemplars and skills (Kuhn, 1970, 1977; Polanyi, 1958). The *axiom view* answers that a theory answers questions posed in a given language and is a logical theory; the *raw-data view*, that it predicts observations and is a program. We assess the axiom view against the theorems of this paper, and say which of our claims are theorems (proofs in Appendix A) and which are arguments.

1.1 An argument map

We separate five claims (Table 1). **D**: a theory answers questions posed in a given language rather than predicting raw data, read comparatively, as a step away from raw-data prediction rather than the end point. **S**: a theory is a logical theory, not a program. **E**: theories explain propositions. **L**: the ladder of **??** (raw data; questions posed with existing abstractions; theories that extend the language; theories that revise the base abstractions). **P**: SAI is progress over raw-data prediction, both in what a theory does and in what it is.

What rests on theorems. Proved, in the stated settings: the idealised formalism cannot see the difference between a theory and a program (Section 1.4); answering questions differs from raw-data prediction in target, data type, coherence and partiality, not in the inference principle (Section 1.3); with a cost on the written theory, usefulness depends exponentially on the pair (reasoner, vocabulary), and on declarative against procedural form only by a constant factor once intermediate concepts are available and the reasoner can chain through them (in the model of **??**; open beyond it); theoretical vocabulary is dispensable in the ideal mixture, while relations or entities pay for finite theories and under charged deduction (**??**); and "Kepler's laws are only approximately right" is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view, since in M_A an approximately true law is a law with a calibrated error model (**??**). Computed, conditional on the pool of forms and the codes: on noise-free interval

data, simplicity among compatible forms is relative to the language, and with graded fit the data decide (**??**). A theorem plus an argued identification: Hempel's inductive-systematisation reply fails in the idealised setting (**??**). Arguments only: the descriptive observation; why declarative shape is good (open); that D survives the objection from statistical practice; the readings of the toys and the general claims about explanation; that the axiom view misses the procedural, statistical and dynamic sides of theories (**??**); and our refusal, resting on a possibility result (**??**), to say anything realist about induced theories.

The ladder. We accept L. The base language of step 2 may be highly theoretical (heliocentric longitudes presuppose Copernican geometry); no theory-neutral observation language is assumed, so the classical objections to one do not apply in the same way. But theory-ladenness (Hanson, 1958) biases comparisons between theories that disagree about the base language, since a theory that revises it is scored in a language it rejects; this is why the ladder has a step 4.

1.2 The descriptive observation and the explanatory question

Claim S combines a description with an explanation, which we keep apart. *The observation:* textbook presentations of fundamental theories are declarative. They state laws, definitions, principles and models given by equations, not procedures that print predictions; this needs no theory of why it holds. Two qualifications. *Declarative is not sentential:* physics texts write something closer to a class of models specified by equations (Suppes, 1960, 1962) than to first-order axioms. We use sentences because questions and data are sentences and the prior of SAI is over texts; the question-answering view itself is neutral between sentences and classes of models, since in partition semantics a question is a partition of logical space (cf. Groenendijk and Stokhof, 1984). *Theories in use have procedural parts:* methods attached to concepts, numerical schemes, simulation codes; in quantum chemistry a "level of theory" is a computational method, and one position of Mars from Kepler's laws already needs root-finding for Kepler's equation. So the observation concerns textbook presentations, not theories in all their uses.

The question, why the declarative shape is good and whether an ideal inducer should produce it, is open. Craig's construction rules out an answer based only on what can be represented (Section 1.4), so an answer must be relative to a reasoner with costs, or to a community of reasoners. In the model of **??**, the best formal answer we have, what matters is a reasoner that can do algebra or chain through intermediate concepts, together with those concepts, not logical form as such. The explanation may lie in communication and piecemeal checking, which have no formal setting yet.

Question 1.1 (Why declarative shape?)**.** Is there a formal setting for a community of bounded reasoners who communicate, check and extend theories piece by piece, in which theories of declarative shape provably do better than programs?

A natural argument for S is that a theory answers in many directions (Kepler's third law gives the period from the semi-major axis and conversely; **??**), while a program computes one direction cheaply. The second half holds of particular programs, not of programs as a class: a generic solver applied to a declarative law costs $O(1)$ extra length, a program with internal variables answers all m directions of a parity law on m variables with $O(m)$ operations, and under program length directions cost nothing. They matter only for a fixed weak reasoner and a cost on the written theory: for one parity law, forward chaining needs exactly m times as many rules as unit propagation needs clauses, over the base variables or with one round, and at most 9 times

the written length of the least clausal theory with intermediate concepts and unbounded rounds (**????**).

1.3 What a theory does: questions against raw data

Questions, and the answers a theory proves, are defined in **??**, following the logic of questions (cf. Belnap and Steel, 1976; Hintikka, 1999); $[k] := \{1, \dots, k\}$.

Definition 1.2 (Answers)**.** A theory T *answers* a question $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ if $B \cup T \vdash \varphi_i$ for some i, and *answers it partially* if $B \cup T \vdash \bigvee_{i \in J} \varphi_i$ for some J with $\emptyset \neq J \subsetneq [k]$.

In the Kepler example (**??**) a question is a cover by interval sentences ("where is Mars at time t, to within $2'$?") or structural ("is the orbit of Mars planar?"); without the elements of the orbits, Kepler's laws answer structural questions but entail no direction datum with tolerance below π (**??**). A raw-data predictor answers one question, "what is the next observation?", and answers it fully. SAI answers every question the base language can pose, and reports how much of its posterior answers none:

Proposition 1.3 (The answer distribution)**.** *Let $M_A(D) > 0$, let $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ be a question, and let $\mathrm{Un}_D(\mathcal{Q}) := \mathbb{U}(E_D \setminus \bigcup_i O_{\varphi_i})/\mathbb{U}(E_D)$, the posterior probability that the random theory proves no direct answer. (a) $\sum_{i \le k} \mathrm{Bel}_D(\varphi_i) + \mathrm{Un}_D(\mathcal{Q}) = 1$. (b) For $\emptyset \neq J \subsetneq [k]$, $\mathrm{Bel}_D(\bigvee_{i \in J} \varphi_i) \geq \sum_{i \in J} \mathrm{Bel}_D(\varphi_i)$, and the difference is the posterior probability that the random theory answers $\mathcal{Q}$ partially through J but proves no φ_i with $i \in J$; for $J = [k]$, $\mathrm{Bel}_D(\bigvee_i \varphi_i) = 1$ and the difference is $\mathrm{Un}_D(\mathcal{Q})$. (c) For $\mathcal{Q} = \{\varphi, \neg\varphi\}$, (a) is the trichotomy $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi) + \mathrm{Indep}_D(\varphi) = 1$ of* **??**.

On E_D the random theory is consistent, so it proves at most one direct answer (Appendix A.1). In SOI the analogue of Un_D mixes logical independence with a labeller's silence (**??**).

The conservation-law world. What is a law worth to each task? In a toy world (Proposition A.1 in Appendix A.2) a micro-state $x \in \{0,1\}^m$ moves by a uniformly random permutation of each weight class, so its weight is conserved; the raw data are the trajectory, and the questions are "what is $|x_t|$?". Knowing the law saves a Bayes predictor of the raw data between $\frac{1}{2}\log\frac{\pi m}{2} + o(1)$ bits and one bit more per step (for $t \le Z/2$, Z the size of the weight class), a share $O(\log m/m)$ of its step loss, until the trajectory returns to its start; on $t_{\max} + 1$ questions it saves the share $t_{\max}/(t_{\max}+1)$ of the no-law loss. Yet by the chain rule the raw saving at each step equals the loss of the raw no-law predictor's own answer to "what is $|x_{t+1}|$?", and in each task a Bayes mixture of the two hypotheses pays at most one bit for learning the law. A computation (code in the accompanying repository) gives savings of 2.348, 3.331 and 5.326 bits at the start of a trajectory at $m = 16$, 64, 1024, against $\frac{1}{2}\log\frac{\pi m}{2} = 2.326$, 3.326, 5.326, and shares 0.15, 0.052, 0.0052. So the share of the raw loss that the law saves tends to 0, because of micro-detail that no theory addresses, while its absolute gain per step equals its value for the corresponding question, and both tasks learn the law at the same price: the theory is "in" the raw predictor. The tasks differ in their *target*, which losses are counted, not in what must be learned nor in what the law is worth per step; scoring questions is right when what we want is answers. (The saving on questions holds by construction: the toy asks only about the conserved quantity.)

Answers never cost more description than raw modelling plus an *extractor* that reads them off the raw data. Let $\mathbb{X}$ be a finite alphabet, μ a computable probability measure on $\mathbb{X}^{\mathbb{N}}$ (the raw world), $(\varphi_q)_{q \in \mathbb{N}}$ a fixed computable family of sentences (the questions), and η a partial computable function with $\eta(q, x) \in \{0, 1\}$ computed from a finite prefix of x; put $f_x(\varphi_q) := \eta(q, x)$ and $\mu_\eta(D) := \mu\{x : f_x \supseteq D\}$.

Proposition 1.4 (Raw modelling plus an extractor)**.** *If f_x is consistent with B for μ-almost every x, then $M_A(D) \geq 2^{-K(\mu)-K(\eta)-c}\mu_\eta(D)$ for every finite data set D of question–answer pairs, so $L_A(D) \leq -\log \mu_\eta(D) + K(\mu) + K(\eta) + c$; here c depends also on the family (φ_q). For a computable sequence x with f_x consistent with B, $L_A(D) \leq K(x) + K(\eta) + c$ on every D with $f_x \supseteq D$.*

A theory machine samples the raw world and emits the extractor's labels (Appendix A.3). The extractor is the bridge from raw data to the question language, and its cost $K(\eta)$ is not small in real cases: it contains whatever recognises a planet in an image. Answering is also the stronger achievement (Proposition A.3 in Appendix A.4): a consistent r.e. theory that proves the data bounds the loss of M_A by its complexity, but on the lottery stream M_A predicts well while no consistent r.e. theory proves the data; predicting data that present an incomplete theory leaves its open questions open; and the beliefs of SAI are monotone under entailment and closed under conjunction up to additive error, unlike, in the semimeasure convention, the predictions of consistent SOI (**??**).

The objection from statistical practice. Much of physics confronts theory with near-raw data through generative models and information-theoretic losses: gravitational-wave searches match waveform templates against the strain series with a Gaussian noise likelihood, cosmology fits likelihoods of microwave-background data, particle physics simulates detectors event by event. D survives in comparative form, we think, because these pipelines exist to answer a few questions (a mass, a spin, a cosmological parameter, whether a particle exists). But the generative model is theory plus instrument plus noise, and the proof-requiring core of SAI, with no native noise model, sits far from this practice (**??**). So D holds in comparative form; its formal content is a change of data type, logical coherence, partiality and the choice of target, not a new inference principle (one-tape SAI $\asymp$ one-tape consistent SOI, **??**); and the real difficulty, the extractor, SAI takes as given.

Purpose and logarithmic loss. Grading a prediction for use and scoring credences are different jobs. For the first, the purpose decides what is graded and with which tolerance, as with the interval data of **??**; Solomonoff axiom induction lets the purpose fix the propositions, and scores credences on those. For the second, the logarithmic score is, up to positive affine transformation, the only strictly proper local scoring rule on more than two outcomes (a result attributed to Bernardo, 1979), but which proper score is privileged is contested (accuracy-first arguments favour quadratic inaccuracy, Leitgeb and Pettigrew, 2010; Pettigrew, 2016), and for bounded or misspecified models different proper losses select different best models. We do not claim that the logarithmic score is privileged; even inside SAI the price of a datum can depend on the loss convention by an unbounded amount (**??**).

1.4 What a theory is, after Craig

The idealised formalism cannot see shape. Every consistent assigner of truth values is an axiom set at $O(1)$ extra bits. Coin-free SAI $\asymp$ coin-free consistent SOI, by Craig's construction inside the universal prior (**??**), and one-tape SAI $\asymp$ one-tape consistent SOI (**??**). Adding vocabulary changes the weights of old-vocabulary data by $O(1)$ (**????**); coin-free, r.e. axiom induction is finite-theory induction in a three-sorted extension, up to $O(1)$ (**??**). Program length sees neither the reasoner nor the vocabulary (**??**); the posterior cannot move between deductively equivalent formulations, whose complexities in the flagpole toys below differ by $O(1)$ however many laws are added (Proposition A.5); and the diagonal separations from SOI under time bounds come

from free deduction, not from axioms (**??**). So the idealised formalism cannot see the difference between a theory and a program. Nor does D support S as a claim about what an ideal inducer should produce: the labellers of SOI carry no theory and are equivalent to theories, and Khardon and Roth (1997) give a learning-to-reason algorithm for a class of propositional languages that has no efficient reasoning algorithm when represented as a formula-based knowledge base.

Where shape matters, it is relative to a reasoner and a vocabulary, and only when the written theory is charged: in the model of **??** the exponential separations are between pairs (reasoner, vocabulary) (**??**); in the settings of **??** what pays is theoretical relations and entities, not vocabulary (**??**). We still find it plausible that good theories are shaped like logical theories; what is missing is a formal setting in which the shape matters (Question 1.1), and nothing in the formal results favours sentences over classes of models. Since the limit posterior can live on random theories, and in true arithmetic with true but adversarially chosen data on false ones (**????**; a possibility result), we say nothing realist about induced theories.

1.5 Explanation

In the axiom view T explains φ when $B \cup T \vdash \varphi$, the deductive-nomological pattern (Hempel and Oppenheim, 1948), and an explanation is better the more a short theory derives, as in unification accounts (Friedman, 1974; Kitcher, 1981); the explanatory power of **??** makes this precise (**??** relates it to Myrvold's account, Myrvold, 2003). But derivability is symmetric where explanation is not: a pole's height and the sun's elevation entail the shadow's length, and the shadow's length and the elevation entail the height (Bromberger's example, Bromberger, 1966, as attributed by Glymour, 1978). Over the theory B of real closed fields, with poles p_i, heights $\mathrm{ht}(p)$ and shadow lengths $\mathrm{sh}(p,u)$ when the tangent of the sun's elevation is $u > 0$, the shadow law has two formulations: Λ_h says $\mathrm{sh}(p,u) \cdot u = \mathrm{ht}(p)$; Λ_s says $\mathrm{sh}(p,u) \cdot u = \mathrm{sh}(p,1)$ and $\mathrm{ht}(p) = \mathrm{sh}(p,1)$. With initial conditions on the heights or on the reference shadows $\mathrm{sh}(p_i,1)$ they give T_H, T_S (with Λ_h) and T'_H, T'_S (with Λ_s). A derivation is charged by the number of closed instances of the law's universally quantified conjuncts that it uses; C_H, C_S extend T_H, T_S by intervention vocabulary with the height, resp. the reference shadow, exogenous (precise forms in Appendix A.5).

Proposition 1.5 (The flagpole)**.** *(a) T_H, T_S, T'_H, T'_S are deductively equivalent over B; so every data set is compatible with all four or with none, in the coin-free version the posterior odds of any two programs for them equal their prior odds on every data set compatible with them, and $|K(T_S) - K(T_H)| \leq c$. (b) Under derivation charges, every shadow datum $\mathrm{sh}(p_i,u) = v$ with $u \neq 1$ that they entail needs exactly one instance from T_H and T'_S, whose initial conditions sit at the hub of their formulation, and exactly two from T_S and T'_H: which quantity is the hub is a property of the formulation, not of the world. (c) C_H and C_S have the same consequences in the old vocabulary, and one interventional datum separates them.*

What the toys show. The explanation relation of SAI is direction-free, and its posterior cannot move between deductively equivalent theories. Kitcher's unificationist answer (Kitcher, 1989), in its usual summary, is that the best systematisation of all our beliefs derives the dimensions of objects from their origin and development; a height-from-shadow pattern would add a pattern, which fails for objects that cast no shadow. With k further laws in which the height figures (Proposition A.5 in Appendix A.6), the formulation with the shadow as hub is longer by an amount growing with k in a literal code without definitions (relative to the primitives, or because some objects cast no shadow), by a constant with one definition, and by $O(1)$ under the universal prior; under derivation charges only a mixed theory loses. So Kitcher's answer is an effect of

whole-theory simplicity that SAI reproduces only under a listing code, like the textbook point of **??**. Direction enters cleanly with interventional questions, once the base language can pose them; but the intervention semantics was written into C_H and C_S by hand, so the toy is consistent with the interventionist account (Woodward, 2003) and gives it no support over rivals such as the causal-process account (Salmon, 1984).

We conclude that derivation from a short theory is explanation in the unification sense, with at least four known gaps; we do not claim that derivation is explanation outright. The gaps are *asymmetry*, as above; *explanation by approximate correction*, since Newton's theory does not entail Kepler's laws but, in an exact toy, proves the honest old law on every window and refutes the exact one (**??**), so that "T explains φ" must include "T derives an honest approximation of φ with an error bound on a domain"; *irrelevance*, as in the standard examples of hexed salt and of a man taking birth-control pills, where the length prior penalises an irrelevant axiom in a whole theory but a derivation does not mark which axioms it used; and the *conjunction problem* (L is derived from $L \wedge L'$), which the length prior and the explanatory power address in part. Unification can register an explanatory gain without a gain in accuracy: against Ptolemy's system with the coincidences of retrogression and opposition stated as axioms, which has the same fit, Copernicus's wins at every steepness exactly when the heliocentric reorganisation costs fewer bits than the coincidences (**??**).

1.6 What separates SAI from raw-data prediction

We replace claim P by the formal differences between SAI and a predictor of an observation stream, sorted into advantages for claims D and E, neutral differences and liabilities (Table 2).What does not separate the two is the inference principle, a Bayes mixture over a universal class, and, in the idealised setting, the shape of hypotheses. This is the progress claim we believe: for what a theory does, SAI is a step in the right direction; for what a theory is, the idealised formalism is neutral.

A Proofs for what a theory does and is

This appendix proves the propositions of Section 1, and states in full the three that the section summarises in prose (Propositions A.1, A.3 and A.5). Appendices A.1 to A.4 concern questions against raw data; Appendices A.5 and A.6 give the precise setting of the flagpole toys, prove Proposition 1.5, and extend it to a system of laws.

A.1 The answer distribution

Proof of Proposition 1.3. On E_D the theory $B \cup A_\omega$ is consistent. If it proved φ_i and φ_j with $i \neq j$, it would prove $\varphi_i \wedge \varphi_j$, while $B \vdash \neg(\varphi_i \wedge \varphi_j)$; so it proves at most one direct answer. Hence the sets $E_D \cap O_{\varphi_i}$, $i \leq k$, are pairwise disjoint, and together with $E_D \setminus \bigcup_i O_{\varphi_i}$ they partition E_D. Dividing the measures by $\mathbb{U}(E_D) > 0$ gives (a), since $\mathrm{Bel}_D(\varphi_i) = \mathbb{U}(E_D \cap O_{\varphi_i})/\mathbb{U}(E_D)$.

(b) For $i \in J$, $O_{\varphi_i} \subseteq O_{\bigvee_{j\in J}\varphi_j}$, and the sets $E_D \cap O_{\varphi_i}$ are pairwise disjoint; so

$$\mathbb{U}\Big(E_D \cap O_{\bigvee_{j\in J}\varphi_j}\Big) = \sum_{i\in J} \mathbb{U}(E_D \cap O_{\varphi_i}) + \mathbb{U}\Big(E_D \cap O_{\bigvee_{j\in J}\varphi_j} \setminus \bigcup_{i\in J} O_{\varphi_i}\Big).$$

Divide by $\mathbb{U}(E_D)$: the last term is the posterior probability that the random theory proves $\bigvee_{j\in J}\varphi_j$, that is, answers $\mathcal{Q}$ partially through J, but proves no φ_i with $i \in J$. For $J = [k]$, $B \vdash \bigvee_i \varphi_i$, so

	difference	support
Advantages for claims D and E		
1	*Data type:* labelled sentences of a chosen language; questions are sets of sentences.	**??** and Def. 1.2
2	*Logical coherence:* consequences of the data get posterior probability 1, while in the semimeasure convention consistent SOI is never certain of an unasked consequence; Bel is monotone under entailment and closed under conjunction up to additive error.	**??** and Prop. A.3
3	*Partiality:* "independent" is a native third value, whose mass is the semimeasure defect; for a k-way question the answer masses and the unanswered mass sum to 1.	**??** and Prop. 1.3
4	*An observed "independent" label* separates theories from labellers by an unbounded amount on one datum, coin-free and one-tape, when $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and $\mathsf{Q} \subseteq \mathrm{Cn}(B)$.	**??**
5	*Laws are sentences with belief values:* $\mathrm{Bel}_D(\forall x\, \varphi)$ differs from the probability that all future data conform.	**??**
6	*Answers cost no more description than raw modelling plus an extractor,* and can cost far less.	Props. 1.4 and A.1
Neutral differences		
7	*Piecemeal confirmation:* credence in single laws, though only whole theories meet the data: the standard Bayesian treatment of Duhem's problem (Dorling, 1979), which Strevens (2001) argues begs the question. SAI inherits the debate; its allocation of blame is fixed by the prior, hence by the syntax.	definitions; an argument
8	*Truth from unprovability:* if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, belief or independent mass on a Π_1 sentence certifies its truth; a curiosity of arithmetic data.	**??**
9	*Explanatory direction needs questions the language can pose:* interventional data separate causal theories that the old vocabulary cannot.	Prop. 1.5(c); in general an argument
Liabilities		
10	*Random theories, false theories:* on the lottery stream bounded one-tape loss is achieved by random theories and by no consistent r.e. theory; in true arithmetic with true (adversarial) data the limit posterior can live on false theories, a possibility result.	**????**
11	*Language-relative simplicity, within bounds:* faithful translations change weights by $2^{\pm(K(\tau)+c)}$; within that bound the base language decides real choices, such as the preferred orbit of Mars on noise-free interval data.	**??**; **??**

Table 2: What formally separates SAI from raw-data prediction. Items 7 and 9 and the sorting are arguments; the other items are results.

$E_D \subseteq O_{\bigvee_i \varphi_i}$ and $\mathrm{Bel}_D(\bigvee_i \varphi_i) = 1$; by (a) the difference is $1 - \sum_i \mathrm{Bel}_D(\varphi_i) = \mathrm{Un}_D(\mathcal{Q})$.

(c) For $\mathcal{Q} = \{\varphi, \neg\varphi\}$ ($k = 2$), $\mathrm{Bel}_D(\neg\varphi) = \mathrm{Dis}_D(\varphi)$ and $\mathrm{Un}_D(\mathcal{Q})$ is the posterior probability that the random theory decides neither, which is $\mathrm{Indep}_D(\varphi)$. □

A.2 The conservation-law world

For even m, micro-states are $x \in \{0,1\}^m$ with weight $|x|$; the dynamics σ is a uniformly random permutation of each weight class, so the weight is conserved (the law); x_0 is uniform on the class of weight $m/2$, of size $Z := \binom{m}{m/2}$, and $x_{t+1} := \sigma(x_t)$. The raw predictors Raw^+, Raw^- are Bayes for "σ is a uniformly random permutation of each weight class", resp. "of $\{0,1\}^m$", and their step-t loss is that of predicting x_{t+1} from $x_0, \ldots, x_t$. The questions are "what is $|x_t|$?", $t = 0, 1, \ldots$; the question predictors Qa^+, Qa^-, $\mathsf{Qa}^{\mathrm{bin}}$ are Bayes for "$|x_t| = |x_0|$, with $|x_0|$ uniform on $\{0, \ldots, m\}$", "the $|x_t|$ are i.i.d. uniform on $\{0, \ldots, m\}$" and "the $|x_t|$ are i.i.d. binomial$(m, \frac{1}{2})$". In each task Mix gives prior $\frac{1}{2}$ to each of the task's two hypotheses ($\mathsf{Raw}^{\pm}$, resp. $\mathsf{Qa}^{\pm}$). Let t^* be the length of the cycle of σ through x_0.

Proposition A.1 (The conservation-law world)**.** *(a) For $t < t^*$ the step-t losses of Raw^+ and Raw^- are $\log(Z-t)$ and $\log(2^m-t)$; for $t \geq t^*$ both are 0. The law's gain $\kappa_t := \log \frac{2^m-t}{Z-t}$ $(t < t^*)$ is nondecreasing in t; for $t \leq Z/2$, $\log \frac{2^m}{Z} \leq \kappa_t \leq \log \frac{2^m}{Z} + 1$, where $\log \frac{2^m}{Z} = \frac{1}{2} \log \frac{\pi m}{2} + o(1)$, and the share $\kappa_t / \log(2^m - t)$ of the step loss is $O(\log m/m)$. (b) Qa^+ pays $\log(m+1)$ for $|x_0|$ and 0 afterwards, Qa^- pays $\log(m+1)$ per question; over $t_{\max}+1$ questions the law's share of the no-law loss is $t_{\max}/(t_{\max}+1)$. (c) In each task the cumulative loss of Mix exceeds that of the better of its two hypotheses by at most one bit, on every trajectory. (d) For $t < t^*$, $\log(2^m - t) = \kappa_t + \log(Z-t)$ with $\kappa_t = -\log P_{\mathsf{Raw}^-}(|x_{t+1}| = m/2 \mid x_0, \ldots, x_t)$: the law's raw gain is exactly the loss of Raw^-'s own answer to "what is $|x_{t+1}|$?". The loss of $\mathsf{Qa}^{\mathrm{bin}}$ on each question is κ_0.*

Lemma A.2 (Next state of a random permutation)**.** *Let σ be a uniformly random permutation of a finite set X, and let $x_0, \ldots, x_t \in X$ be distinct with $\sigma(x_s) = x_{s+1}$ for $s < t$. Conditionally on these t values of σ, $\sigma(x_t)$ is uniform on $(X \setminus \{x_0, \ldots, x_t\}) \cup \{x_0\}$, a set of $|X| - t$ elements.*

Proof. Conditionally on $\sigma(x_s) = x_{s+1}$ for $s < t$, the restriction of σ to $X \setminus \{x_0, \ldots, x_{t-1}\}$ is a uniformly random bijection onto $X \setminus \{x_1, \ldots, x_t\}$, since every such bijection extends the given values to exactly one permutation. So $\sigma(x_t)$ is uniform on $X \setminus \{x_1, \ldots, x_t\} = (X \setminus \{x_0, \ldots, x_t\}) \cup \{x_0\}$, which has $|X| - t$ elements. □

Proof of Proposition A.1. (a) For $t < t^*$ the states $x_0, \ldots, x_t$ are distinct, all of weight $m/2$, and x_{t+1} is either unvisited or x_0. Under the hypothesis of Raw^+ the permutations of the weight classes are independent, and the trajectory stays in the class of x_0; Lemma A.2 with X that class ($|X| = Z$) gives the predictive probability $1/(Z-t)$ of the true x_{t+1}, a step loss of $\log(Z-t)$. Under the hypothesis of Raw^- the lemma with $X = \{0,1\}^m$ gives $\log(2^m - t)$. For $t \geq t^*$ the path has returned to x_0, the cycle through x_0 is known to both predictors, and later states are determined: both losses are 0.

Since $2^m > Z \geq t^* > t$, the map $t \mapsto (2^m - t)/(Z-t)$ is increasing on $[0, Z)$ (its derivative is $(2^m - Z)/(Z-t)^2 > 0$), so κ_t is nondecreasing and $\kappa_t \geq \kappa_0 = \log(2^m/Z)$. For $t \leq Z/2$, $(2^m - t)/(Z-t) \leq 2^m/(Z/2)$, so $\kappa_t \leq \log(2^m/Z) + 1$. By Stirling's formula, $Z = \binom{m}{m/2} = 2^m \sqrt{2/(\pi m)}\,(1 + o(1))$, so $\log(2^m/Z) = \frac{1}{2} \log \frac{\pi m}{2} + o(1)$. Finally $\kappa_t \leq \log(2^m/Z) + 1 = O(\log m)$, and $\log(2^m - t) \geq \log(2^m - Z/2) \geq m - 1$ since $Z \leq 2^m$; so the share is $O(\log m/m)$ for $t \leq Z/2$.

(b) Qa^+ predicts $|x_0|$ uniformly on $\{0, \ldots, m\}$, a loss of $\log(m+1)$, and then $|x_t| = |x_0|$ with probability 1, which is true since the weight is conserved: loss 0. Qa^- pays $\log(m+1)$ on each question. Over the questions $t = 0, \ldots, t_{\max}$ the losses are $\log(m+1)$ and $(t_{\max}+1)\log(m+1)$, and the law saves the share $t_{\max}/(t_{\max}+1)$ of the latter.

(c) In each task Mix is the Bayes mixture with prior $\frac{1}{2}$ on the task's two hypotheses; x_0 has the same law under both raw hypotheses, so the posterior after x_0 is still $(\frac{1}{2}, \frac{1}{2})$. By the chain rule, the cumulative loss of a Bayes mixture is $-\log$ of the mixture probability of the whole sequence. If the two hypotheses give the sequence probabilities 2^{-a} and 2^{-b}, the mixture's loss is $-\log(\frac{1}{2}2^{-a} + \frac{1}{2}2^{-b}) \leq -\log(\frac{1}{2}2^{-\min(a,b)}) = \min(a,b) + 1$, on every trajectory.

(d) Under Raw^-, by Lemma A.2, x_{t+1} is uniform on a set of $2^m - t$ states, of which exactly $Z - t$ have weight $m/2$: the $Z-(t+1)$ unvisited states of the class of x_0, and x_0. So $P_{\mathsf{Raw}^-}(|x_{t+1}| = m/2 \mid x_0, \ldots, x_t) = (Z-t)/(2^m - t) = 2^{-\kappa_t}$, and conditionally on that weight x_{t+1} is uniform on those $Z - t$ states, which is exactly the predictive distribution of Raw^+. The chain rule splits the step loss $\log(2^m - t)$ of Raw^- into κ_t, the loss of its prediction of the weight, and $\log(Z-t)$, the loss of Raw^+. Finally, every $|x_t|$ equals $m/2$, and the binomial probability of $m/2$ is $Z2^{-m}$, so $\mathsf{Qa}^{\mathrm{bin}}$ pays $\log(2^m/Z) = \kappa_0$ on each question. □

Computations. A computation (code in the accompanying repository, seeded) gives the following. At the start of a trajectory the law's gain per raw step is 1.871, 2.148, 2.348, 2.837, 3.331, 3.829, 5.326 and 10.292 bits at $m = 8$, 12, 16, 32, 64, 128, 1024 and 10^6, against $\frac{1}{2}\log\frac{\pi m}{2} = 1.826$, 2.118, 2.326, 2.826, 3.326, 3.826, 5.326, 10.292; its share of the raw step loss is 0.23, 0.18, 0.15, 0.089, 0.052, 0.030, 0.0052 and 1.0×10^{-5}. The value $\kappa_{Z/2}$ is 2.659, 2.976, 3.199, 3.732, 4.258, 4.777 and 6.308 bits at $m = 8, \ldots, 1024$; the loss of Qa^- per question is 3.170, 3.700, 4.088, 5.044, 6.022, 7.011 and 10.001 bits, about twice κ_0. At $m = 12$ ($Z = 924$), averaging exactly over the cycle length (uniform on $\{1, \ldots, Z\}$), the expected raw loss over 300 steps is 2415.6 bits with the law and 3004.9 without (share 0.196), and the mixture's is 2416.6; over the 301 questions the losses are 3.70 and 1113.8 bits (share 0.997). Over a whole cycle the raw share is 0.232 and the mean gain 2.75 bits per step. In 2000 simulated runs the mean raw losses are within one standard error of these values, and the mixture never exceeds the better hypothesis by more than one bit. The identity of (d) holds at $m = 12$ for every $t < Z$ to within 1.8×10^{-15} bits in floating point.

A.3 Questions from raw data and an extractor

Proof of Proposition 1.4. Sampling. There is a partial computable map Ψ from tapes to $\mathbb{X}^{\mathbb{N}}$, computable from an index of μ, that is defined for $\mathbb{U}$-almost every tape and satisfies $\Psi_*\mathbb{U} = \mu$. This is the standard construction: read tape bits to refine a dyadic interval, and output the next symbol s as soon as the interval lies inside the μ-interval of s given the symbols already output; it fails only on the null set of tapes that hit interval boundaries. Each output symbol depends on a finite prefix of the tape.

The theory machine. Let G be the theory machine that, on a tape ω, computes $\Psi(\omega)$ as far as needed and, for each q, emits $\varphi_q^{\eta(q,\Psi(\omega))}$ once the tape prefix read so far fixes a prefix of $\Psi(\omega)$ on which the computation of $\eta(q,\cdot)$ halts. Each emission depends on a finite prefix of ω, so G is a c.e. set of pairs (prefix, sentence) as **??** requires, and $K(G) \le K(\mu) + K(\eta) + c$, where c includes an index of the fixed family (φ_q). If $\Psi(\omega)$ is defined, then $A_\omega^G = \Gamma_{f_{\Psi(\omega)}}$, the set of labelled sentences of the labelling $f_{\Psi(\omega)}$.

The bound. Let H be the set of x for which f_x is consistent with B; $\mu(H) = 1$ by hypothesis. On every tape ω with $\Psi(\omega)$ defined, $\Psi(\omega) \in H$ and $f_{\Psi(\omega)} \supseteq D$, the theory $B \cup A_\omega^G$ is consistent and proves ψ_D, so $\omega \in E_D^G$. Hence $\nu_G(D) \ge \mathbb{U}\{\omega : \Psi(\omega) \in H,\ f_{\Psi(\omega)} \supseteq D\} = \mu\{x \in H : f_x \supseteq D\} = \mu_\eta(D)$. By **??**, $M_A(D) \ge 2^{-K(\nu_G)}\nu_G(D) \ge 2^{-K(G)}\nu_G(D) \ge 2^{-K(\mu)-K(\eta)-c}\mu_\eta(D)$, and taking $-\log$ gives the bound on L_A. For a computable sequence x, take $\mu = \delta_x$, with $K(\delta_x) \le K(x) + c$; the hypothesis is that f_x is consistent with B, and $\mu_\eta(D) = 1$ exactly when $f_x \supseteq D$. □

The constant depends on the question family because the theory machine must write the sentences φ_q. The converse of the proposition fails badly, since answers do not determine raw data; in the conservation-law world the raw loss exceeds the question loss by the whole micro-entropy (Proposition A.1).

A.4 Answering and predicting

The following gives a formal vocabulary for "answering questions is more than predicting data": answering implies predicting, (a), but not conversely, (b), and success at predicting data that present a theory does not settle the questions the data leave open, (c). Part (d) is the coherence that a predictor of separate labels lacks.

Proposition A.3 (Answering and predicting)**.** *(a) If a consistent r.e. theory T proves every datum of a consistent reveal sequence, the cumulative loss of M_A is at most $-\log \mathbb{U}_A(\mathrm{Ext}T) \le K_{\equiv}(T)+c$ in the semimeasure convention, in any order. (b) On the lottery stream of* **??***, M_A has bounded loss in the semimeasure convention, while no consistent r.e. theory proves every datum and $W_A(D_n) \to 0$. (c) On two-valued data presenting an incomplete consistent r.e. theory T^*, the limit posterior is the prior restricted to the consistent extensions of T^*, and on every sentence independent of $B \cup T^*$ all three values* Bel, Dis, Indep *have positive limits; three-valued data identify the class of T^*. (d) For all D with $M_A(D) > 0$ and all sentences,* $\mathrm{Bel}_D(\bigwedge_{i\le k}\varphi_i) \ge 1-\sum_{i\le k}(1-\mathrm{Bel}_D(\varphi_i))$*, and* $\mathrm{Bel}_D(\psi) \ge \mathrm{Bel}_D(\varphi)$ *whenever $B \cup \{\varphi\} \vdash \psi$. So the sentences with belief at least $1-\delta$ are closed under entailment and, with error $k\delta$, under k-fold conjunction.*

Proof of Proposition A.3. (a) By **??**(d) the cumulative loss in the semimeasure convention is $L_A(D_n) - L_A(\emptyset)$, in any order, and by **??** it is at most $-\log \mathbb{U}_A(\mathrm{Ext}T) \le K_{\equiv}(T) + c$.

(b) In the setting of **??**, part (a) of that theorem bounds $M_A(D_n)$ below by a positive constant independent of n, so the cumulative loss $L_A(D_n) - L_A(\emptyset)$ is bounded; part (b) says that no consistent r.e. theory proves every datum (with B); and part (c) gives $W_A(D_n) \to 0$.

(c) This is **??**(b), (c): if $B \cup T^*$ is incomplete and $\mathrm{Cn}(D^{\pm}_{\infty}) = \mathrm{Cn}(T^*)$, the limit posterior is the prior restricted to the consistent extensions of T^*, and for every ψ independent of $B \cup T^*$ the limits of $\mathrm{Bel}_{D_n}(\psi)$ and $\mathrm{Dis}_{D_n}(\psi)$ lie in $(0,1)$ while $\lim_n \mathrm{Indep}_{D_n}(\psi) \ge \mathbb{U}_A[T^*]/\mathbb{U}_A(\mathrm{Ext}T^*) > 0$; and if every sentence appears with its T^*-trichotomy label, the posterior probability of the class of T^* tends to 1.

(d) A theory that proves φ and ψ proves $\varphi \wedge \psi$, so $O_\varphi \cap O_\psi \subseteq O_{\varphi\wedge\psi}$. Within E_D,

$$\mathbb{U}(O_\varphi \cap O_\psi \cap E_D) \ge \mathbb{U}(O_\varphi \cap E_D) + \mathbb{U}(O_\psi \cap E_D) - \mathbb{U}(E_D);$$

dividing by $\mathbb{U}(E_D)$ gives $\mathrm{Bel}_D(\varphi\wedge\psi) \ge \mathrm{Bel}_D(\varphi) + \mathrm{Bel}_D(\psi) - 1$, and induction on k gives $\mathrm{Bel}_D(\bigwedge_{i\le k}\varphi_i) \ge 1 - \sum_{i\le k}(1-\mathrm{Bel}_D(\varphi_i))$. If $B \cup \{\varphi\} \vdash \psi$, then $O_\varphi \subseteq O_\psi$, so $\mathrm{Bel}_D(\psi) \ge \mathrm{Bel}_D(\varphi)$. □

A.5 The flagpole

Setting. The language has two sorts, poles and reals; B is the theory of real closed fields on the real sort. There are pole constants $p_1, \dots, p_{N_\mathrm{p}}$, a function symbol ht from poles to reals (the height) and a function symbol sh from poles and reals to reals ($\mathrm{sh}(p,u)$ is the length of the shadow of p when the tangent of the sun's elevation is $u > 0$). Data are sentences $\mathrm{sh}(p_i,u) = v$ and $\mathrm{ht}(p_i) = v$ with rational numerals u, v; interval versions behave in the same way. The two formulations of the shadow law are

$$\Lambda_h := \forall p\,\forall u\,(u > 0 \to \mathrm{sh}(p,u)\cdot u = \mathrm{ht}(p)),$$
$$\Lambda_s := \forall p\,\forall u\,(u > 0 \to \mathrm{sh}(p,u)\cdot u = \mathrm{sh}(p,1)) \ \wedge\ \forall p\,\mathrm{ht}(p) = \mathrm{sh}(p,1),$$

and, with rational numerals a_i for the heights, the four theories are

$$T_H := \Lambda_h + \{\mathrm{ht}(p_i) = a_i\}_{i\le N_\mathrm{p}}, \qquad T_S := \Lambda_h + \{\mathrm{sh}(p_i,1) = a_i\}_{i\le N_\mathrm{p}},$$
$$T'_H := \Lambda_s + \{\mathrm{ht}(p_i) = a_i\}_{i\le N_\mathrm{p}}, \qquad T'_S := \Lambda_s + \{\mathrm{sh}(p_i,1) = a_i\}_{i\le N_\mathrm{p}}.$$

In T_S the shadow at 45° "explains" the height. The *intervention vocabulary* consists of $\mathrm{sh}_H(p,u,v)$, the shadow length after the height of p is set to v, and $\mathrm{ht}_S(p,v)$, the height after the reference

shadow $\mathrm{sh}(p,1)$ is set to v by an intervention on the shadow (for instance, by an obstacle). The causal extensions, with the height exogenous and with the shadow exogenous, are

$$C_H := T_H + \{\forall p\,\forall u\,\forall v\,(u>0 \to \mathrm{sh}_H(p,u,v)\cdot u = v),\ \forall p\,\forall v\,\mathrm{ht}_S(p,v) = \mathrm{ht}(p)\},$$
$$C_S := T_S + \{\forall p\,\forall u\,\forall v\,\mathrm{sh}_H(p,u,v) = \mathrm{sh}(p,u),\ \forall p\,\forall v\,\mathrm{ht}_S(p,v) = v\}.$$

Derivation charges. The cost of a derivation is the number of closed instances it uses of the universally quantified conjuncts of the law axioms (Λ_h has one such conjunct, Λ_s two); steps in B and uses of the initial conditions are free. The *least cost* of a datum for a theory is the least number of such instances that, together with B and the theory's initial conditions, entail it. By completeness this is the least cost of a derivation, so the count is semantic, as in Herbrand's theorem.

The precise form of Proposition 1.5(b) is as follows. For a shadow datum $\mathrm{sh}(p_i,u) = a_i/u$ with $u \neq 1$, the least costs for T_H, T_S, T'_H, T'_S are 1, 2, 2, 1; for a height datum $\mathrm{ht}(p_i) = a_i$ they are 0, 1, 0, 1; and for a reference-shadow datum $\mathrm{sh}(p_i,1) = a_i$ they are 1, 0, 1, 0. So the two hub theories T_H and T'_S have equal costs on all shadow data with $u \neq 1$. Their law texts differ by the constant $|\Lambda_s| - |\Lambda_h|$, and in a literal code (Appendix A.6) their initial conditions differ by $N_{\mathrm{p}}\Delta_{\mathrm{sh}}$, where $\Delta_{\mathrm{sh}} := |\mathrm{sh}(x,1)| - |\mathrm{ht}(x)|$ is the extra length of the reference-shadow term; this is 0 if the language has a primitive for the reference shadow, and then the two texts differ by $|\Lambda_s| - |\Lambda_h|$, independently of N_{p} and the data. The precise form of (c) is: C_H and C_S have the same consequences in the language without sh_H and ht_S; $C_H \vdash \mathrm{sh}_H(p_1,1,2a_1) = 2a_1$ ("doubling the pole doubles its 45° shadow"), while $C_S \vdash \mathrm{sh}_H(p_1,1,2a_1) = a_1$, which refutes it when $a_1 \neq 0$; and $C_H \vdash \mathrm{ht}_S(p_1,v) = a_1$ while $C_S \vdash \mathrm{ht}_S(p_1,v) = v$, for every numeral v.

Lemma A.4 (Least costs in a star)**.** *Let T be one of the theories of this appendix without the dark poles: law axioms whose universally quantified conjuncts have closed instances of the form $\pi \to s_1 = \rho\cdot s_2$, with π a condition on real numerals, s_1, s_2 terms of one pole and ρ a real term that B proves nonzero when π holds, and one initial condition $s(p_i) = a_i$ per pole. For a pole p_i, let the* instance graph *have as vertices the closed terms $\mathrm{ht}(p_i)$, $\mathrm{sh}(p_i,u)$ (terms whose real arguments B proves equal are identified) and, in Appendix A.6, $\mathrm{qt}_j(p_i)$, and an edge for each closed instance whose condition B proves, joining the two terms it relates (an instance that relates a term to itself is trivial and is ignored). If the instance graph is a tree, the least cost of a datum $s'(p_i) = v$ that T entails is the distance in the graph from $s'(p_i)$ to the vertex fixed by the initial condition.*

Proof. The instances along the path from the fixed vertex to $s'(p_i)$, together with B and the initial condition, determine the value of $s'(p_i)$; since T entails the datum, that value is v, so the distance is an upper bound. Conversely, let I be a set of fewer instances than the distance. As the graph is a tree, the edges of I do not connect $s'(p_i)$ to the fixed vertex. Fix a model of T over $\mathbb{R}$ (each theory of this appendix has one). Build a model of B, the initial conditions and I over $\mathbb{R}$ as follows: on the connected component (for the edges of I) of the fixed vertex, and on every term of another pole, keep the values of the fixed model; on the component of $s'(p_i)$, give $s'(p_i)$ a value other than v and propagate it along the edges of I, which is consistent because the component is a tree and each edge is an invertible linear relation; give all other terms arbitrary values. Instances with a false condition hold trivially. This model falsifies the datum, so I does not entail it. □

Proof of Proposition 1.5. (a) At $u = 1$, Λ_h gives $\mathrm{sh}(p,1) = \mathrm{ht}(p)$; given this, $\mathrm{sh}(p,u)\cdot u = \mathrm{ht}(p)$ and $\mathrm{sh}(p,u)\cdot u = \mathrm{sh}(p,1)$ are equivalent. So $B \vdash \Lambda_h \to \Lambda_s$; conversely Λ_s gives $\mathrm{ht}(p) = \mathrm{sh}(p,1)$ and

hence Λ_h. Given either law, the initial conditions $\mathrm{ht}(p_i) = a_i$ and $\mathrm{sh}(p_i, 1) = a_i$ are interderivable, so the four theories are deductively equivalent over B. Compatibility of a theory with a data set (consistent with B and proving every datum) depends only on its deductive closure, so every data set is compatible with all four or with none. In the coin-free version the posterior weight of a program is $2^{-|p|}$ times the indicator of compatibility, divided by the normaliser, so the odds of two compatible programs are their prior odds. The map on axiom lists that replaces each $\mathrm{ht}(p_i) = a_i$ by $\mathrm{sh}(p_i, 1) = a_i$, and its inverse, are fixed computable maps, so $|K(T_S) - K(T_H)| \le c$.

(b) For each pole, the instances of Λ_h join $\mathrm{ht}(p_i)$ to each $\mathrm{sh}(p_i, u)$ (a star centred at the height), and the instances of Λ_s join $\mathrm{sh}(p_i, 1)$ to each $\mathrm{sh}(p_i, u)$ with $u \neq 1$ and to $\mathrm{ht}(p_i)$ (a star centred at the reference shadow); instances with $u \le 0$ have false conditions, and instances at other poles are not connected to p_i. Both graphs are trees, and Lemma A.4 gives the counts: in the height-centred star with the height fixed (T_H), shadows and the reference shadow are at distance 1; with the reference shadow fixed (T_S), a shadow with $u \neq 1$ is at distance 2 and the height at 1; in the shadow-centred star with the height fixed (T'_H), a shadow with $u \neq 1$ is at distance 2 and the reference shadow at 1; with the reference shadow fixed (T'_S), shadows and the height are at distance 1. The law texts of T_H and T'_S are Λ_h and Λ_s; their initial conditions differ only in the terms $\mathrm{ht}(p_i)$ and $\mathrm{sh}(p_i, 1)$, once per pole.

(c) Without the intervention symbols, C_H and C_S are T_H and T_S plus axioms about fresh symbols. Every model of T_H expands to a model of C_H (put $\mathrm{sh}_H(p, u, v) := v/u$ for $u > 0$, arbitrary otherwise, and $\mathrm{ht}_S(p, v) := \mathrm{ht}(p)$), and every model of T_S to a model of C_S (put $\mathrm{sh}_H(p, u, v) := \mathrm{sh}(p, u)$ and $\mathrm{ht}_S(p, v) := v$). So C_H and C_S are conservative over T_H and T_S, whose consequences coincide by (a). The interventional consequences are instances of the added axioms: in C_H, $\mathrm{sh}_H(p_1, 1, 2a_1)\cdot 1 = 2a_1$ and $\mathrm{ht}_S(p_1, v) = \mathrm{ht}(p_1) = a_1$; in C_S, $\mathrm{sh}_H(p_1, 1, 2a_1) = \mathrm{sh}(p_1, 1) = a_1$ and $\mathrm{ht}_S(p_1, v) = v$. If $a_1 \neq 0$ then $2a_1 \neq a_1$, so C_S refutes the datum $\mathrm{sh}_H(p_1, 1, 2a_1) = 2a_1$ that C_H proves. □

Computations. A computation (code in the accompanying repository) confirms the counts by exhaustive search over all subsets of finite instance pools, with exact linear algebra over the rationals, on 500 random data sets with one to four poles (shadows at $u = 2$, and at $u = 3$ in a second implementation); on 2000 random data the four theories prove, refute or leave open exactly the same data. In the literal syntax of that code (numerals counted as one token), $|\Lambda_s| - |\Lambda_h| = 20$ tokens and $\Delta_{\mathrm{sh}} = 1$ token, so T_S is longer than T_H by one token per pole.

A.6 The flagpole with a system of laws

Kitcher's answer to the asymmetry appeals to other laws in which the height figures, which the toy of Proposition 1.5 lacks. We add such laws.

Setting. As in Appendix A.5, with further function symbols $\mathrm{qt}_1, \ldots, \mathrm{qt}_k$ from poles to reals (further quantities: weight, the height on the construction plan, and so on), nonzero rational numerals $\theta_1, \ldots, \theta_k$, and data $\mathrm{qt}_j(p_i) = v$. The laws in height form and in shadow form are $\Lambda_j^{\mathrm{ht}} := \forall p\, \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)$ and $\Lambda_j^{\mathrm{sh}} := \forall p\, \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{sh}(p, 1)$. The theories are the height hub $T_H^k := \Lambda_h + \{\Lambda_j^{\mathrm{ht}}\}_{j \le k} + \{\mathrm{ht}(p_i) = a_i\}_i$, the mixed theory $T_S^k := \Lambda_h + \{\Lambda_j^{\mathrm{ht}}\}_j + \{\mathrm{sh}(p_i, 1) = a_i\}_i$, and the shadow hub $T_{SS}^k := \Lambda_s + \{\Lambda_j^{\mathrm{sh}}\}_j + \{\mathrm{sh}(p_i, 1) = a_i\}_i$.

Dark variant. Add a predicate Sunlit on poles, restrict the shadow law to sunlit poles, $\Lambda_h^{\mathrm{D}} := \forall p\, (\mathrm{Sunlit}(p) \to \forall u\, (u > 0 \to \mathrm{sh}(p, u)\cdot u = \mathrm{ht}(p)))$, and take sunlit poles $p_1, \ldots, p_{N_{\mathrm{p}}}$ and dark poles

$p'_1, \ldots, p'_{N_\mathrm{d}}$, with the axioms $\mathrm{Sunlit}(p_i)$ and $\neg\mathrm{Sunlit}(p'_l)$ in every theory and rational numerals a'_l for the heights of the dark poles. Laws in shadow form must split cases:

$$\Lambda_j^{\mathrm{sh,D}} := \forall p \Big(\big(\mathrm{Sunlit}(p) \to \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{sh}(p,1)\big) \wedge \big(\neg\mathrm{Sunlit}(p) \to \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)\big) \Big).$$

The theories are $T_H^{k,\mathrm{D}} := \Lambda_h^{\mathrm{D}} + \{\Lambda_j^{\mathrm{ht}}\}_j + \{\mathrm{ht}(p_i) = a_i\}_i + \{\mathrm{ht}(p'_l) = a'_l\}_l$ and $T_{SS}^{k,\mathrm{D}} := \Lambda_h^{\mathrm{D}} + \{\Lambda_j^{\mathrm{sh,D}}\}_j + \{\mathrm{sh}(p_i,1) = a_i\}_i + \{\mathrm{ht}(p'_l) = a'_l\}_l$, each with the Sunlit axioms. A *literal code* charges a theory by the number of symbols it writes, each axiom separately, without definitions; $|\cdot|$ denotes this length, a symbol qt_j counts as one token, and $\Delta_{\mathrm{sh}} := |\mathrm{sh}(x,1)| - |\mathrm{ht}(x)|$ is the extra length of the reference-shadow term ($\Delta_{\mathrm{sh}} = 0$ if the language has a primitive for the reference shadow).

Proposition A.5 (The flagpole with a system of laws)**.** *(a)* Class tie. *T_H^k, T_S^k and T_{SS}^k are deductively equivalent over B, and so are $T_H^{k,\mathrm{D}}$ and $T_{SS}^{k,\mathrm{D}}$; so for every k no data set separates them, and their deductive classes coincide. (b)* Universal prior. *$|K(T_{SS}^k) - K(T_H^k)| \le c$ and $|K(T_{SS}^{k,\mathrm{D}}) - K(T_H^{k,\mathrm{D}})| \le c$, with c independent of k, N_p, N_d and the numerals. (c)* Literal codes. *$|T_{SS}^k| - |T_H^k| = (|\Lambda_s| - |\Lambda_h|) + (k + N_\mathrm{p})\Delta_{\mathrm{sh}}$. In the dark variant, $|T_{SS}^{k,\mathrm{D}}| - |T_H^{k,\mathrm{D}}| = k\,\Delta_{\mathrm{split}} + \sum_{j\le k}|\theta_j| + N_\mathrm{p}\Delta_{\mathrm{sh}}$, where $\Delta_{\mathrm{split}} > 0$ is the length of the case-split frame and $|\theta_j|$ the length of the numeral θ_j, which the split law writes twice; so the shadow hub loses by at least $k\,\Delta_{\mathrm{split}}$, an amount growing with k, even when $\Delta_{\mathrm{sh}} = 0$. With one definition the overhead of the laws is a constant: define $\mathrm{z}(p)$ once as $\mathrm{sh}(p,1)$ on sunlit poles and $\mathrm{ht}(p)$ on dark ones, and write every law in* z*; in every model of the theory,* z *is coextensive with* ht*. (d)* Derivation charges *(instances of universally quantified conjuncts, as in Appendix A.5). A datum $\mathrm{qt}_j(p_i) = \theta_j a_i$ has least cost* 1 *for T_H^k and T_{SS}^k, and* 2 *for the mixed theory T_S^k. On shadow data with $u \ne 1$ the two hub theories have least cost* 1 *each; on heights T_H^k has* 0 *and T_{SS}^k has* 1*; on reference shadows the reverse.*

Proof. (a) Each theory contains a form of the shadow law, which gives $B \vdash \mathrm{ht}(p) = \mathrm{sh}(p,1)$ for every pole p (for every sunlit pole in the dark variant), as in the proof of Proposition 1.5(a). Substituting, each law in shadow form is equivalent to the law in height form, and each reference-shadow initial condition to the corresponding height condition. In the dark variant, the case split collapses to $\mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)$, because on sunlit poles $\mathrm{sh}(p,1) = \mathrm{ht}(p)$ by Λ_h^{D} at $u = 1$.

(b) The syntactic translations and their inverses are fixed computable maps on axiom lists: from T_H^k to T_{SS}^k, replace Λ_h by Λ_s, each Λ_j^{ht} by Λ_j^{sh} and each $\mathrm{ht}(p_i) = a_i$ by $\mathrm{sh}(p_i,1) = a_i$; from $T_H^{k,\mathrm{D}}$ to $T_{SS}^{k,\mathrm{D}}$, replace each Λ_j^{ht} by $\Lambda_j^{\mathrm{sh,D}}$ (reading θ_j and qt_j off the axiom) and the initial condition of each pole declared sunlit by its reference-shadow form. So a program for one list yields a program for the other at a constant extra length, independent of k, N_p, N_d and the numerals.

(c) Counting symbols: each Λ_j^{sh} differs from Λ_j^{ht} by one occurrence of the replaced term, and so does each sunlit initial condition; this gives $(k + N_\mathrm{p})\Delta_{\mathrm{sh}}$ besides the law texts. Each $\Lambda_j^{\mathrm{sh,D}}$ is longer than Λ_j^{ht} by a fixed frame of length $\Delta_{\mathrm{split}} > 0$ (the connectives, the two Sunlit atoms, a second copy of $\mathrm{qt}_j(p) = \cdot$, and the term $\mathrm{sh}(p,1)$) and by a second copy of the numeral θ_j; both dark theories contain Λ_h^{D}, the dark initial conditions and the Sunlit axioms. The definition of z is a single axiom of constant length, and laws written in z have the length of the laws written in ht. In every model of the theory, $\mathrm{z}(p) = \mathrm{sh}(p,1) = \mathrm{ht}(p)$ on sunlit poles by Λ_h^{D} at $u = 1$, and $\mathrm{z}(p) = \mathrm{ht}(p)$ on dark poles by the definition.

(d) For each pole the instance graphs are trees: for T_H^k and T_S^k, a star centred at $\mathrm{ht}(p_i)$ with leaves $\mathrm{sh}(p_i,u)$ and $\mathrm{qt}_j(p_i)$; for T_{SS}^k, a star centred at $\mathrm{sh}(p_i,1)$ with leaves $\mathrm{sh}(p_i,u)$ ($u \ne 1$), $\mathrm{ht}(p_i)$ and $\mathrm{qt}_j(p_i)$. Lemma A.4 gives the counts: $\mathrm{qt}_j(p_i)$ is at distance 1 from the fixed centre in T_H^k and

T^k_{SS}, and at distance 2 from the fixed leaf $\mathrm{sh}(p_i, 1)$ in T^k_S; the shadow, height and reference-shadow counts are those of Proposition 1.5(b) for T_H and T'_S. □

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